

MANILA, Philippines

WMO: 984250

Lat: 14.583N

Lon: 120.983E

Elev: 13

StdP: 101.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	23.8	17.3	12.4	26.6	18.1	13.0	26.5	6.4	28.6	5.8	28.9	2.3	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.5	34.5	26.4	33.8	26.4	33.2	26.3	28.2	31.8	27.8	31.4	27.6	31.1	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	30.4	26.9	22.6	30.3	26.6	22.1	30.1	90.8	31.7	89.2	31.6	87.7	31.2	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
7.9	6.4	5.5	DB	20.7	36.3	1.9	1.1	19.3	37.1	18.2	37.7	17.1	38.3	15.6	39.0	
			WB	18.8	29.0	1.2	0.7	17.9	29.5	17.2	29.9	16.5	30.3	15.7	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	27.0	27.5	28.6	30.1	30.3	29.5	28.5	28.4	28.4	28.3	27.5
		DBStd	1.53	1.15	1.15	1.19	1.15	1.30	1.37	1.27	1.20	1.15	1.05	1.20
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0
		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0
		CDD10.0	6776	526	491	577	603	629	584	575	572	552	575	542
		CDD18.3	3734	267	258	318	353	371	334	317	313	302	317	284
		CDH23.3	44396	2568	2676	3762	4723	5048	4326	3765	3703	3535	3807	2981
(14)	Wind	WSAvg	2.8	2.4	2.5	2.6	2.7	2.9	3.0	3.2	3.3	3.0	2.6	2.5
		PrecAvg	2134	10	4	12	23	126	312	438	462	358	202	133
		PrecMax	3338	39	18	45	67	469	725	1744	1188	885	411	352
		PrecMin	1531	0	0	0	0	15	96	100	211	36	31	0
		PrecStd	487	11	6	16	26	126	194	400	236	225	108	98
		DB	31.7	32.5	33.7	35.2	35.3	34.9	33.5	33.1	32.7	33.0	32.9	32.3
		MCWB	24.4	24.5	25.4	25.9	26.4	27.4	27.1	27.1	26.7	26.1	25.4	24.9
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	30.7	31.6	32.8	34.3	34.5	33.7	32.4	32.1	32.0	32.1	32.0	31.3
		MCWB	24.3	24.4	25.2	25.8	26.6	27.1	26.9	27.1	26.8	26.1	25.7	24.9
		DB	30.0	30.9	32.1	33.7	33.8	32.9	31.7	31.4	31.4	31.5	31.3	30.5
		MCWB	24.2	24.3	25.1	25.7	26.7	27.0	26.9	26.9	26.7	26.1	25.6	24.9
		DB	29.3	30.1	31.4	32.9	33.1	32.1	31.0	30.8	30.8	30.9	30.7	29.8
		MCWB	24.0	24.2	24.9	25.8	26.8	27.0	26.7	26.7	26.5	26.0	25.5	24.7
		WB	26.1	26.6	27.7	28.1	28.6	28.7	28.6	28.1	28.0	28.0	27.6	26.7
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.7	29.9	30.7	32.4	32.6	32.9	31.6	30.8	30.9	31.3	30.8	30.4
		WB	25.6	25.9	26.6	27.3	28.1	28.1	27.7	27.6	27.4	27.2	26.9	26.2
		MCDB	29.1	29.7	30.6	31.9	32.2	31.9	30.8	30.6	30.5	30.5	30.2	29.7
		WB	25.1	25.4	26.1	26.9	27.7	27.7	27.3	27.3	27.1	26.9	26.5	25.7
		MCDB	28.7	29.2	30.5	31.6	31.9	31.3	30.5	30.3	30.3	30.1	29.9	29.1
		WB	24.7	24.9	25.6	26.5	27.3	27.3	27.0	27.0	26.7	26.6	26.1	25.3
		MCDB	28.3	28.8	30.1	31.3	31.5	30.9	30.3	30.1	29.9	29.7	29.6	28.6
(35)	Mean Daily Temperature Range	MDBR	4.6	5.1	5.5	5.9	5.5	5.0	4.5	4.1	4.4	4.5	4.6	4.5
		5% DB	5.3	6.0	6.3	6.6	6.5	5.9	5.5	5.2	5.3	5.3	5.4	5.3
		MCWBR	2.5	2.6	2.7	2.7	2.7	2.4	2.4	2.3	2.3	2.3	2.3	2.3
		5% WB	4.7	5.1	5.7	5.9	5.6	5.3	4.9	4.6	4.8	4.7	4.8	4.7
		MCWBR	2.8	2.8	3.0	2.9	2.7	2.6	2.5	2.4	2.4	2.4	2.6	2.5
		taub	0.421	0.419	0.451	0.477	0.476	0.468	0.464	0.472	0.468	0.478	0.439	0.440
		taud	2.418	2.408	2.321	2.253	2.286	2.323	2.335	2.304	2.311	2.262	2.388	2.370
(42)	Clear-Sky Solar Irradiance	Ebn at Noon	874	891	869	840	828	826	831	833	841	825	849	843
		Edh at Noon	114	119	133	142	135	129	128	133	132	136	116	117
		RadAvg	4.23	5.00	5.65	6.33	5.73	5.14	4.53	4.37	4.64	4.47	4.19	3.78
(45)	All-Sky Solar Radiation	RadStd	0.25	0.43	0.38	0.34	0.51	0.52	0.58	0.45	0.47	0.37	0.37	0.32

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.32	+0.47	N/A	+0.45	+0.40	+0.54	N/A	N/A	+158	+158

Nomenclature: See separate page